

CS 663: Digital Image Processing

Assignment 1 Report

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1 Image Subsampling and Interpolation

1.1 1(a) Image Shrinking and Moiré Effects

Implementation

The image shrinking algorithm was implemented to reduce the spatial resolution of the input image `suit.png` by a factor of d along each dimension. The function `myImageShrink()` was developed to perform this subsampling by selecting every d -th pixel along the rows and columns. Assuming a 1:1 aspect ratio for the axes, the new dimensions are calculated simply as the original width and height divided by d . For any given coordinate (x, y) in the newly shrunk image, the corresponding pixel intensity is sampled directly from the original image at the coordinate $(x \cdot d, y \cdot d)$.

Observations and Aliasing Analysis

The input image (`suit.png`) contains dense, high-frequency spatial details, specifically the fine, repetitive woven stripes of the jacket fabric. The assignment required generating subsampled images for $d = 2$ and $d = 3$ to demonstrate Moiré effects.

When the image is subsampled without applying a prior low-pass anti-aliasing filter, the sampling rate falls below the Nyquist rate required to accurately capture the fabric's dense grid. Consequently, high-frequency spatial information aliases into lower frequencies. This violation produces stark Moiré patterns—highly visible, artificial wavy curves and swirling interference bands that do not exist in the original pattern.

As the subsampling factor increases from $d = 2$ to $d = 3$, the sampling rate drops further below the Nyquist threshold, increasing the severity of the data loss and drastically altering the geometry and thickness of the Moiré interference bands. As requested, the pixel units are displayed along each axis for all output images.

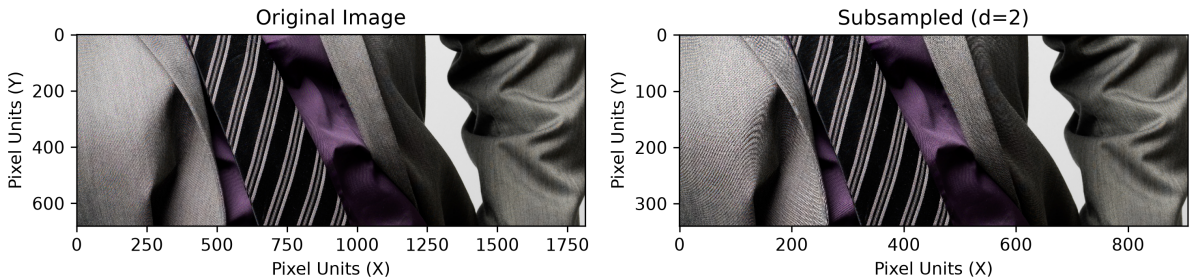


Figure 1: Comparison of the original image and the $d = 2$ subsampled output. Artificial Moiré interference patterns appear on the fabric due to aliasing.

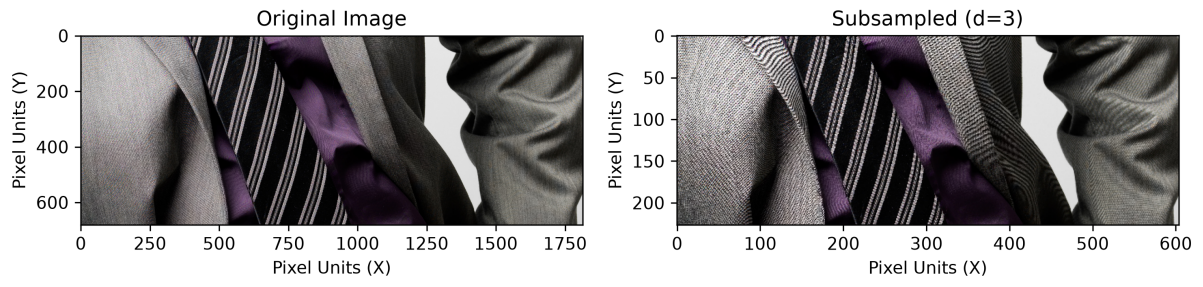


Figure 2: Comparison of the original image and the $d = 3$ subsampled output. The Moiré effects become significantly more pronounced and distorted as the subsampling factor increases.